considerably. Follow the instructions on the mkcert GitHub page to install it, and then run

```
mkcert -install
to generate the CA certificate and install it. The CA cert
will automatically be marked as trusted by web browsers
installed on your operating system.
```

> **DEFINITION** A ***self-signed certificate*** is a certificate that has been signed using the private key associated with that same certificate, rather than by a trusted certificate authority. Self-signed certificates should be used only when you have a direct trust relationship with the certificate owner, such as when you generated the certificate yourself.

You can now generate a certificate for your Spark server running on localhost. By default, `mkcert` generates certificates in Privacy Enhanced Mail (PEM) format. For Java, you need the certificate in PKCS#12 format, so run the following command in the root folder of the Natter project to generate a certificate for localhost:

```
mkcert -pkcs12 localhost
```

The certificate and private key will be generated in a file called `localhost.p12`. By default, the password for this file is `changeit`. You can now enable HTTPS support in Spark by adding a call to the `secure()` static method, as shown in listing 3.4. The first two arguments to the method give the name of the keystore file containing the server certificate